

# Nhat Hung Nguyen

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## EDUCATION

### Bachelor of Applied Science and Engineering (B.A.Sc) in Electrical Engineering

Toronto Metropolitan University (Formerly known as Ryerson University)

Sep. 2021 - May 2026

- Relevant courses: Electronic Circuits I/II, Software Systems, Electric and Magnetic Fields, Microprocessor Systems, Signal and System, Energy Conservation, Power Electronics, Power System Analysis, Advanced Electric Drives, Low Power Digital Integrated Circuit

## SKILLS

**Programming/Simulation** || AutoCAD, Python, Java, KiCAD, MATLAB, C, LTSpice, Arduino, VHDL/Verilog, Cadence (Virtuoso), Altium Designer, Revit, Quartus

**Tools** || Oscilloscope, Function Generator, Power Supply, Signal Generator, Control System, BJT, Soldering, DC Power Supplies, Printed Circuit Board(PCB) Schematic Design, ESP32, ESP8266, OP-AMPS, FPGA, MOSFETs, RTL Designs

**Other** || Microsoft Office [Teams, Word, Publisher, Powerpoint, Excel], Analytical, Organization, Communication, Teamwork, Multitasking, Enthusiastic, Problem-solving, Leadership, Time Management, **Fluent in English/Vietnamese**

## WORK EXPERIENCE

### Maintenance Worker | CRM Rubber Manufacturers - Brantford, ON

May 2026 - Present

- Conducted routine hourly inspections of production machinery to verify normal operating conditions.
- Prepared operational reports and communicated equipment irregularities to management to support preventive maintenance activities.
- Supported troubleshooting efforts by observing system performance and reporting irregularities.

### Senior Designer Of Ryerson Electronic Chapter - Toronto, ON

Sept. 2022 – May. 2024

- Applications: C/C++, Soldering, Oscilloscope, Microcontroller, Arduino, ESP32, DMM
- Developed a functional RC car prototype by programming an Arduino UNO and an ESP32 microcontroller in C/C++ and designing its electronic layout.
- Designed and soldered custom control circuits, integrating motor drivers and wireless modules to create a responsive user control system.
- Conducted verification work on the physical and interactive design through testing, troubleshooting hardware and software integration using an oscilloscope and digital multimeter to ensure a reliable user experience.

## RELEVANT PROJECT

### Smart Bartender Ordering System | Python, FastAPI, ESP8266, Arduino, REST APIs, HTML/CSS/JavaScript, Wi-Fi

- Designed and implemented a full-stack automated beverage dispensing system integrating a FastAPI (Python) backend, ESP8266 microcontroller, and web-based user interface.
- Developed RESTful APIs and JSON-based Wi-Fi communication to manage drink orders, system state, and synchronized software-to-hardware execution.
- Programmed ESP8266 firmware (Arduino C/C++) to parse backend instructions and translate drink recipes into timed relay control signals for pump-based liquid dispensing.
- Implemented a machine learning recommendation model that analyzes historical user preferences to suggest personalized beverages.

### Multi-Stage Analog Circuit Design and Waveform Generator | BJT, OP-AMPS, MOSFETS, Function Generator

- Developed, simulated, analyzed, implemented, and tested a transistor amplifier with multiple stages operating on a single supply, ensuring it met a predefined set of specifications.
- Conceived, simulated, executed, and validated a Linear Voltage-Controlled Multi-function Waveform Generator (VCFG) circuit - primarily focused on delivering a practical demonstration of knowledge through waveform.

### VHDL/Verilog-Based Arithmetic Processor Design | Verilog, Quartus, VHDL, RTL Design

- Designed and implemented a custom arithmetic processor featuring an ALU and control unit using VHDL and Verilog.